

Inferential Persistence under Drift

Geometric Laws for Temporal Validity

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Abstract

The object of study is validity of an information state under drift, with temporal validity as one distinguished slice. An information state may be a windowed empirical law, a transport estimator, a calibration set, or a controller memory, but in every case retained data reduce finite-sample noise while accumulating mismatch with the present. For an information-state pipeline ϕ and a flow family τ , the validity functional $V(\phi, \tau)$ records the best achievable present-time error over memory scale. In the deterministic H^{older}–Wasserstein model, this yields the constitutive law

$$\mathbb{E} d(\hat{P}_t^{(n)}, P_t) \leq C_K n^{-a} + C_S \zeta n^H,$$

with horizon scale $n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}$ and a non-trivial useful-memory band. A Gaussian location benchmark closes the exponent of the law, and a tractable one-dimensional proof model together with projection and sliced transfer establish the same mechanism beyond the benchmark class. Lag-geometry observability yields plug-in inference and operational identifiability for the horizon. Empirically, the predicted U-curves, roughness-dependent scaling, validity-before-detection gap, and sequential phase map all appear in synthetic and real streams. A split-conformal benchmark under drift exhibits the same geometry: the calibration window has an interior optimum, coverage degrades before alarm, and the empirically safe calibration band coincides with the useful-memory band. Validity is therefore a structural property of retained information itself, with temporal validity and conformal calibration as operational readouts.

Keywords: tracking under drift; validity; finite-memory tracking; distribution tracking; conformal calibration; Wasserstein distance.

MSC 2020: 93D05, 93D30, 68P99.

1 Introduction

Validity under drift operates through an information state built from recent data. A windowed empirical law, a geometric estimator, a conformal calibration set, or a controller memory all compress the past and are then used to estimate, calibrate, or decide in the present. These objects are useful only while the retained information remains valid for the current target.

Write ϕ for a pipeline that retains recent data, updates that retained state, and extracts a present-time readout. Let $\tau = \{\tau_d : d \in \mathcal{D}\}$ be a flow family indexed by dimensions such as time, space, structure, or operation. The validity functional

$$V(\phi, \tau) = \inf_{n \geq 1} (C_K(\phi) n^{-a(\phi)} + C_S \zeta(\tau) n^{H(\tau)})$$

assigns a persistence value to that pipeline against a flow class τ . Here $C_K n^{-a}$ is the finite-sample acquisition term induced by the chosen state geometry, while $C_S \zeta n^H$ is the drift-induced dissipation of that state under the admissible flow. Retention lowers sampling noise and ages the retained information at the same time. Detection and calibration expose the same mechanism: an alarm can remain silent after retained evidence has already become stale, and a calibration set can lose coverage before any detector reacts.

Let P_t denote the present target law, let $\bar{P}_t^{(n)}$ denote the window target formed from the last n target laws, and let $\hat{P}_t^{(n)}$ be the estimator built from the last n observations. In the deterministic H^{older}–Wasserstein constitutive model, tracking error separates into a finite-sample term relative to $\bar{P}_t^{(n)}$ and a staleness term between $\bar{P}_t^{(n)}$ and P_t :

$$\mathbb{E} d(\hat{P}_t^{(n)}, P_t) \leq C_K n^{-a} + C_S \zeta n^H.$$

When the finite-sample term decays like n^{-a} and the staleness term grows like ζn^H , their balance defines the validity horizon and the induced useful-memory region. In the deterministic H^{older}–Wasserstein constitutive model, this balance realizes the validity functional

$$V(\phi, \tau) = \inf_{n \geq 1} (C_K(\phi) n^{-a(\phi)} + C_S \zeta(\tau) n^{H(\tau)}).$$

The finite-sample exponent a depends on the geometry used to compare distributions or evaluate readouts, while the roughness exponent H and amplitude ζ describe how quickly retained information loses validity. The resulting horizon is structural: it is imposed jointly by metric geometry and drift regularity. The temporal coordinate is one slice of this validity surface: if $d = \text{time}$ is one coordinate in $\mathcal{D}$, then $V(\phi, \tau_d)$ recovers validity as a special case.

The law raises three linked tasks: identify the horizon imposed by sampling and drift, infer that horizon from data, and act on it when the memory scale becomes a control variable. The main theorem closes the exponent of the horizon law on a Gaussian location benchmark over a deterministic H^{older} class. A tractable one-dimensional root- n

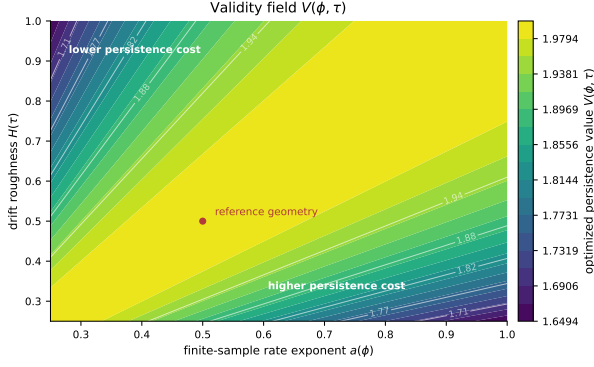


Figure 1: Validity field over the (a, H) plane. Lower-left geometries favor persistence; upper-right geometries incur stronger drift penalty.

proof anchor and projection-based transfer show that the same mechanism persists beyond that benchmark.

Horizon inference becomes possible when lag geometry reveals the drift law. Plug-in asymptotics from lag geometry yield confidence structure for the horizon and an operational identifiability score for the useful-memory band. Sequential control closes the operational layer: once update cost is explicit, the best policy depends on drift roughness and on actuation price, producing a phase map rather than a universal winner.

The same object also organizes calibration under drift. A conformal calibration window is retained information used to deliver present-time coverage. In the split-conformal benchmark reported here, the calibration score exhibits the same U-curve geometry as distribution tracking: small windows are noisy, large windows are stale, coverage degrades before alarm, and the safe calibration band aligns with the useful-memory band. Conformal prediction is therefore one operational readout of the validity field.

2 Temporal-Validity Horizon

2.1 Separability and Horizon Geometry

Lemma 2.1 (Separability of V). *Let d be a probability metric satisfying the triangle inequality. At time t , let P_t be the present target law, let $\bar{P}_t^{(n)}$ be the window target, and let $\hat{P}_t^{(n)}$ be the estimator built from the last n observations. Assume that for some constants $C_K > 0$, $C_S > 0$, exponent $a > 0$, roughness index $H \in (0, 1]$, and amplitude $\zeta > 0$,*

$$\mathbb{E} d(\hat{P}_t^{(n)}, \bar{P}_t^{(n)}) \leq C_K n^{-a}$$

and

$$d(\bar{P}_t^{(n)}, P_t) \leq C_S \zeta n^H$$

hold for all n in the regime of interest. Then

$$\mathbb{E} d(\hat{P}_t^{(n)}, P_t) \leq C_K n^{-a} + C_S \zeta n^H.$$

Proof. By the triangle inequality,

$$d(\hat{P}_t^{(n)}, P_t) \leq d(\hat{P}_t^{(n)}, \bar{P}_t^{(n)}) + d(\bar{P}_t^{(n)}, P_t).$$

Taking expectations and applying the two assumptions yields the claim.

This envelope is the constitutive law of temporal validity for retained information. The path class supplies dissipation, while the chosen information state and sample regime provide finite-sample acquisition. The decomposition itself is elementary; the substantive content lies in the benchmark closure, transfer, lower-theory, and operational consequences that follow.

Theorem 2.2 (Main result: Gaussian location minimax benchmark on the deterministic Hölder class). *Fix $H \in (0, 1]$, $\zeta > 0$, and $\sigma > 0$. Let $\mathfrak{M}_H(\zeta)$ denote the class of deterministic mean paths $(\mu_s)_{s \in \mathbb{Z}}$ such that*

$$|\mu_{t-j} - \mu_t| \leq \zeta j^H \quad \text{for all } j \geq 0.$$

Suppose the observations are independent with

$$Y_{t-j} \sim N(\mu_{t-j}, \sigma^2), \quad j = 0, 1, \dots, n-1,$$

and let the target law at time t be $P_t = N(\mu_t, \sigma^2)$. Define the optimal finite-memory minimax risk

$$\mathcal{R}_H^*(\sigma, \zeta) := \inf_{n \geq 1} \inf_{\hat{P}_t^{(n)}} \sup_{\mu \in \mathfrak{M}_H(\zeta)} \mathbb{E} W_2(\hat{P}_t^{(n)}, P_t).$$

Then there exist constants $0 < c_H \leq C_H < \infty$, depending only on H , such that

$$c_H \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)} \leq \mathcal{R}_H^*(\sigma, \zeta) \leq C_H \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

In particular, the deterministic Hölder class has horizon scale

$$n_H^* \asymp (\sigma/\zeta)^{2/(2H+1)}.$$

Moreover, the uniform-window mean estimator has an exact asymptotic upper constant on this benchmark. If $y_H > 0$ solves

$$Hy_H(2\Phi_G(y_H) - 1) = \varphi_G(y_H),$$

then its asymptotic horizon shape is

$$A_H^{\text{mean}} = ((H+1)y_H)^{2/(2H+1)}$$

and its asymptotic risk constant is

$$U_H^{\text{mean}} = ((H+1)y_H)^{-1/(2H+1)} \times \left(2\varphi_G(y_H) + y_H(2\Phi_G(y_H) - 1) \right).$$

The closed forms above are explicit for the uniform-window mean estimator on the benchmark class. They sharpen the constructive upper side of the benchmark, while the lower side is sharp at the exponent level.

Proposition 2.3 (Triangular-array transfer for Hadamard-differentiable functionals). *Let $\mathcal{T}$ be a functional on probability laws that is Hadamard differentiable at P_t tangentially to a linear subspace of signed measures, with derivative $\tilde{T}_{P_t}$ bounded on the relevant tangent class. Suppose that for the triangular-array estimator and its window target, the base metric satisfies*

$$\mathbb{E} d(\hat{P}_t^{(n)}, \bar{P}_t^{(n)}) \leq C_K n^{-a}$$

and the drift envelope satisfies

$$d(\bar{P}_t^{(n)}, P_t) \leq C_S \zeta n^H.$$

If the remainder in the Hadamard expansion is $o(n^{-a})$ uniformly on the relevant class, then there exist constants $\tilde{C}_K, \tilde{C}_S > 0$ such that

$$\mathbb{E} \|\mathcal{T}(\hat{P}_t^{(n)}) - \mathcal{T}(P_t)\| \leq \tilde{C}_K n^{-a} + \tilde{C}_S \zeta n^H.$$

The transfer is exact at the level of the envelope: the geometry changes, but the finite-sample and drift exponents do not.

Proof. Write the Hadamard expansion at P_t and apply the boundedness of the derivative to the centered triangular-array fluctuation. The drift term contributes through the base-metric envelope because $\mathcal{T}$ is locally Lipschitz on the class. The remainder is lower order by assumption, so the transformed law inherits the horizon envelope.

Proposition 2.4 (Finite-dimensional smooth transfer). *Let $\phi_1, \dots, \phi_k$ be bounded measurable functions and let $Z(P) = (\mathbb{E}_P[\phi_1], \dots, \mathbb{E}_P[\phi_k])$. Let $g \in C^2(U)$ on an open set $U \subset \mathbb{R}^k$ containing the relevant range of Z . Define $T(P) = g(Z(P))$. Suppose that on the triangular-array window target and estimator,*

$$\mathbb{E} \|Z(\hat{P}_t^{(n)}) - Z(\bar{P}_t^{(n)})\| \leq C_K n^{-a}$$

and

$$\|Z(\bar{P}_t^{(n)}) - Z(P_t)\| \leq C_S \zeta n^H.$$

If the Hessian of g is bounded by M on the relevant range, then

$$\mathbb{E} |T(\hat{P}_t^{(n)}) - T(P_t)| \leq \tilde{C}_K n^{-a} + \tilde{C}_S \zeta n^H,$$

and the Taylor remainder is quadratic in the moment error:

$$\begin{aligned} |T(P') - T(P) - \nabla g(Z(P))^\top (Z(P') - Z(P))| \\ \leq \frac{M}{2} \|Z(P') - Z(P)\|^2. \end{aligned}$$

Again, the envelope is preserved at the level of exponents, with the constants absorbed into $\tilde{C}_K$ and $\tilde{C}_S$.

Proof. Apply the multivariate Taylor formula to g around $Z(P_t)$. The bounded Hessian gives the stated quadratic remainder. The first-order term is controlled by the assumed finite-sample and drift envelopes for Z , and the argument from proposition 2.3 yields the horizon law for T .

Corollary 2.5 (Temporal-validity horizon). *Under the assumptions of lemma 2.1, the upper envelope*

$$\Phi(n) := C_K n^{-a} + C_S \zeta n^H$$

is minimized, up to constant factors depending only on a , H , and C_S , at

$$n^*(a, H) \asymp (C_K / \zeta)^{1/(a+H)},$$

and the corresponding error scale is

$$\Phi(n^*) \asymp C_K^{H/(a+H)} \zeta^{a/(a+H)}.$$

Proof. Balance the decreasing finite-sample term against the increasing staleness term:

$$C_K n^{-a} \asymp C_S \zeta n^H.$$

This gives $n^{a+H} \asymp C_K / \zeta$, hence the displayed horizon law. Substituting that scale back into either term gives the optimal error scale.

Write the continuous optimizer of Φ as

$$n_\star := \left(\frac{a C_K}{H C_S \zeta} \right)^{1/(a+H)}.$$

This optimizer differs from the scale $n^*(a, H)$ only by constants depending on a , H , and C_S , so the asymptotic horizon law remains unchanged.

Definition 2.6 (Validity field). For an information-state pipeline ϕ , comprising retention, update, and readout maps, with finite-sample envelope $C_K(\phi) n^{-a(\phi)}$ and a drift class τ with dissipation envelope $C_S \zeta(\tau) n^{H(\tau)}$, define

$$V(\phi, \tau) := \inf_{n \geq 1} (C_K(\phi) n^{-a(\phi)} + C_S \zeta(\tau) n^{H(\tau)}),$$

and at the continuous optimizer,

$$V(\phi, \tau) = \kappa(a, H) C_K(\phi)^{H/(a+H)} (C_S \zeta(\tau))^{a/(a+H)},$$

where $\kappa(a, H) = (a + H) a^{-a/(a+H)} H^{-H/(a+H)}$.

Remark 2.7 (Validity surface). If $\tau = \{\tau_d : d \in \mathcal{D}\}$ is a flow family indexed by dimensions such as time, space, structure, or operation, then $V(\phi, \tau_d)$ defines a validity surface over $\mathcal{N} \times \mathcal{D}$. Temporal validity is the slice $d = \text{time}$.

Conjecture 2.8 (Multiaxis validity surface). *For broad classes of retention pipelines ϕ and flow families τ_d , the surface $V(\phi, \tau_d)$ admits a finite horizon in each coordinate d , with the corresponding useful-memory region measurable by a slice-wise analogue of $\mathcal{U}_\delta$. Temporal validity is the case $d = \text{time}$.*

2.2 Useful-Memory Geometry

The horizon scale induces a near-optimal region of memory lengths. For any $\delta > 0$, define the useful-memory region by

$$\mathcal{U}_\delta := \{n > 0 : \Phi(n) \leq (1 + \delta) \Phi(n_\star)\}.$$

This region records the admissible validity band at a fixed tolerance.

Proposition 2.9 (Useful-memory region at tolerance δ). *Let $\Phi(n) = C_K n^{-a} + C_S \zeta n^H$ with $a, H > 0$, and let $n_\star$ be its continuous optimizer. Then the normalized profile*

$$\Psi(x) := \frac{\Phi(n_\star x)}{\Phi(n_\star)} = \frac{H x^{-a} + a x^H}{a + H}$$

is strictly decreasing on $(0, 1]$ and strictly increasing on $[1, \infty)$. Consequently,

$$\mathcal{U}_\delta = [n_-, n_+] = n_\star [x_-(\delta; a, H), x_+(\delta; a, H)],$$

where $x_-(\delta; a, H) \leq 1 \leq x_+(\delta; a, H)$ are the two solutions of $\Psi(x) = 1 + \delta$. In particular, the useful-memory region scales linearly with the horizon scale $n_\star$, and its relative shape depends only on a , H , and δ .

Proof. Since $n_\star$ minimizes Φ , differentiation gives

$$a C_K n_\star^{-a} = H C_S \zeta n_\star^H.$$

Substituting $n = n_\star x$ into Φ and dividing by $\Phi(n_\star)$ yields the displayed identity for Ψ . Differentiation then gives

$$\Psi'(x) = \frac{aH}{a+H} (x^{H-1} - x^{-a-1}),$$

which is negative for $x < 1$ and positive for $x > 1$. Since $\Psi(1) = 1$ and $\Psi(x) \rightarrow \infty$ as $x \rightarrow 0$ or $x \rightarrow \infty$, the sublevel set $\{x > 0 : \Psi(x) \leq 1 + \delta\}$ is a compact interval containing 1, which proves the representation of $\mathcal{U}_\delta$.

For small tolerance, the useful-memory region has the local expansion

$$x_\pm(\delta; a, H) = 1 \pm \sqrt{\frac{2\delta}{aH}} + o(\sqrt{\delta}), \quad \delta \downarrow 0,$$

because $\Psi''(1) = aH$. The local curvature therefore controls how quickly the near-optimal region contracts around the horizon.

The product aH is the local validity curvature of the normalized useful-memory profile at the optimizer.

The horizon depends jointly on the finite-sample rate exponent and the roughness exponent. Different geometries and different drift regularities induce different memory scales.

2.3 Drift Regularity and Staleness Control

The staleness side of the law is controlled by a deterministic path class. For $H \in (0, 1]$ and $\zeta > 0$, write

$$\mathfrak{P}_H(\zeta) := \{(P_s)_{s \in \mathbb{Z}} : W_2(P_{t-j}, P_t) \leq \zeta j^H \text{ for all } j \geq 0\}.$$

The case $H = 1$ is the local Lipschitz envelope; smaller H corresponds to slower validity dissipation with lag.

Lemma 2.10 (Hölder staleness bound via averaged couplings). *Let*

$$\bar{P}_t^{(n)} := \frac{1}{n} \sum_{j=0}^{n-1} P_{t-j}.$$

If the target path belongs to $\mathfrak{P}_H(\zeta)$, then

$$W_2^2(\bar{P}_t^{(n)}, P_t) \leq \frac{\zeta^2}{n} \sum_{j=0}^{n-1} j^{2H},$$

and therefore, by Jensen's inequality,

$$W_2(\bar{P}_t^{(n)}, P_t) \leq \zeta \left(\frac{1}{n} \sum_{j=0}^{n-1} j^{2H} \right)^{1/2} = C_{H,n} \zeta n^H,$$

where

$$C_{H,n} := \left(n^{-(2H+1)} \sum_{j=0}^{n-1} j^{2H} \right)^{1/2}.$$

In particular, $C_{H,n} \rightarrow (2H+1)^{-1/2}$ as $n \rightarrow \infty$.

Proof. For each lag j , let π_j be an optimal coupling between P_{t-j} and P_t . Their average

$$\bar{\pi} := \frac{1}{n} \sum_{j=0}^{n-1} \pi_j$$

couples $\bar{P}_t^{(n)}$ with P_t , so

$$\begin{aligned} W_2^2(\bar{P}_t^{(n)}, P_t) &\leq \int \|x - y\|^2 d\bar{\pi}(x, y) \\ &= \frac{1}{n} \sum_{j=0}^{n-1} W_2^2(P_{t-j}, P_t). \end{aligned}$$

Applying the Hölder envelope gives the displayed finite- n constant, and taking square roots completes the proof.

Combining lemma 2.10 with corollary 2.5 yields the horizon family

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

In the root- n finite-sample regime $a = 1/2$,

$$n^*(1/2, H) \asymp (C_K/\zeta)^{2/(1+2H)}.$$

The finite- n constant in lemma 2.10 is the constant used here for uniform windows; the simpler $(2H+1)^{-1/2}$ expression is its large-window limit.

2.4 One-Dimensional Proof Anchor

The bounded-support fixed-span one-dimensional triangular array provides a tractable rigorous finite-sample instantiation of the abstract law. The model is built on common compact support, uniformly regular window mixtures, and an integrated Bahadur remainder hypothesis.

Proposition 2.11 (Root- n finite-sample bound in the 1D proof model). *Let $X_{n,1}, \dots, X_{n,n}$ be independent one-dimensional observations with laws $P_{n,1}, \dots, P_{n,n}$, and define the window mixture*

$$\bar{P}_n = \frac{1}{n} \sum_{j=1}^n P_{n,j}.$$

Assume:

- (i) *all $P_{n,j}$ are supported on a common compact interval $[L, U]$;*
- (ii) *the within-window drift span is uniformly bounded in n ;*
- (iii) *each mixture law $\bar{P}_n$ has a density $\bar{f}_n$ on $[L, U]$ that is bounded below by a positive constant and uniformly Hölder;*
- (iv) *writing q_n for the quantile function of $\bar{P}_n$ and $\hat{q}_n$ for the empirical quantile function of $\hat{P}_n^{\text{tri}}$, one has the Bahadur representation*

$$\hat{q}_n(u) - q_n(u) = \frac{u - \hat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))} + R_n(u), \quad u \in (0, 1),$$

with

$$\mathbb{E} \int_0^1 R_n(u)^2 du = o(n^{-1}).$$

Then

$$\mathbb{E} W_2^2(\hat{P}_n^{\text{tri}}, \bar{P}_n) = O(n^{-1}),$$

and therefore, by Jensen's inequality,

$$\mathbb{E} W_2(\hat{P}_n^{\text{tri}}, \bar{P}_n) = O(n^{-1/2}).$$

Proof. In one dimension,

$$W_2^2(\hat{P}_n^{\text{tri}}, \bar{P}_n) = \int_0^1 (\hat{q}_n(u) - q_n(u))^2 du,$$

so the problem reduces to the integrated L_2 error of the empirical quantile process. By item (iv),

$$\hat{q}_n(u) - q_n(u) = \frac{u - \hat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))} + R_n(u).$$

Define

$$Z_n(u) := \frac{u - \hat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))}.$$

Then

$$\mathbb{E} \int_0^1 (\hat{q}_n(u) - q_n(u))^2 du = \mathbb{E} \int_0^1 (Z_n(u) + R_n(u))^2 du.$$

Let $m > 0$ be a uniform lower bound for $\bar{f}_n$. Since

$$\hat{F}_n(x) = \frac{1}{n} \sum_{j=1}^n \mathbf{1}\{X_{n,j} \leq x\},$$

independence across the row gives

$$\text{Var}(\hat{F}_n(q_n(u))) = \frac{1}{n^2} \sum_{j=1}^n p_{n,j}(u)(1 - p_{n,j}(u)) \leq \frac{1}{4n},$$

where $p_{n,j}(u) = P_{n,j}((-\infty, q_n(u)])$. Therefore

$$\mathbb{E} \int_0^1 Z_n(u)^2 du \leq m^{-2} \int_0^1 \text{Var}(\hat{F}_n(q_n(u))) du \leq \frac{1}{4m^2 n}.$$

By item (iv),

$$\mathbb{E} \int_0^1 R_n(u)^2 du = o(n^{-1}).$$

The cross term is then $o(n^{-1})$ by Cauchy–Schwarz, since $\mathbb{E} \int_0^1 Z_n(u)^2 du = O(n^{-1})$ and $\mathbb{E} \int_0^1 R_n(u)^2 du = o(n^{-1})$. Hence the expected squared Wasserstein error is $O(n^{-1})$, and Jensen’s inequality gives the root- n rate for $\mathbb{E} W_2(\hat{P}_n^{\text{tri}}, \bar{P}_n)$.

The main conclusion is the exponent. Under this bounded-support fixed-span regime, the triangular-array finite-sample term remains root- n .

2.5 Verifiable Smooth Deformation Class

An explicit deformation class makes the proof-model hypotheses verifiable. Let

$$g_\delta(x) = x + \delta x(1 - x), \quad x \in [0, 1], \quad |\delta| \leq \Delta < 1.$$

Then g_δ is an increasing C^2 diffeomorphism of $[0, 1]$, with

$$1 - \Delta \leq g'_\delta(x) \leq 1 + \Delta, \quad |g''_\delta(x)| \leq 2\Delta,$$

so the transformed densities satisfy

$$\frac{1}{1 + \Delta} \leq f_\delta(y) \leq \frac{1}{1 - \Delta},$$

and the same support is preserved.

Corollary 2.12 (Verifiable Bahadur remainder for the smooth deformation class). *Assume that each row law has the form*

$$P_{n,j} = (g_{\delta_{n,j}})_\# \text{Unif}[0, 1], \quad |\delta_{n,j}| \leq \Delta < 1,$$

and that the within-window span is uniformly bounded in n . Then item (iv) of proposition 2.11 holds. This is the verified deformation class used in the 1D proof anchor. More precisely, there exists a constant $C_\Delta < \infty$ such that

$$\mathbb{E} \int_0^1 R_n(u)^2 du \leq C_\Delta n^{-2}.$$

Consequently,

$$\mathbb{E} W_2(\hat{P}_n^{\text{tri}}, \bar{P}_n) = O(n^{-1/2})$$

on this deformation class.

Proof. Write $\bar{F}_n$ and q_n for the mixture cdf and quantile. Since each $g_{\delta_{n,j}}$ is an increasing diffeomorphism of $[0, 1]$, every mixture law $\bar{P}_n$ is supported on $[0, 1]$ and has density $\bar{f}_n = \frac{1}{n} \sum_{j=1}^n f_{\delta_{n,j}}$ satisfying the explicit bounds

$$\frac{1}{1 + \Delta} \leq \bar{f}_n(y) \leq \frac{1}{1 - \Delta}, \quad \|\bar{f}'_n\|_\infty \leq \frac{2\Delta}{(1 - \Delta)^3}.$$

Hence the inverse map $\bar{F}_n^{-1}$ is uniformly twice differentiable on $(0, 1)$, with second derivative bounded by a constant depending only on Δ . Applying the second-order Taylor expansion of $\bar{F}_n^{-1}$ around $u = \bar{F}_n(q_n(u))$ gives

$$\hat{q}_n(u) - q_n(u) = \frac{u - \hat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))} + R_n(u),$$

with

$$|R_n(u)| \leq C_\Delta |\hat{F}_n(q_n(u)) - u|^2.$$

Now

$$\hat{F}_n(q_n(u)) = \frac{1}{n} \sum_{j=1}^n \mathbf{1}\{X_{n,j} \leq q_n(u)\}$$

is an average of independent Bernoulli variables with mean u . Uniformly in u , its fourth centered moment is $O(n^{-2})$. Therefore

$$\mathbb{E} R_n(u)^2 \leq C_\Delta \mathbb{E} |\hat{F}_n(q_n(u)) - u|^4 \leq C'_\Delta n^{-2},$$

and integrating over $u \in (0, 1)$ yields

$$\mathbb{E} \int_0^1 R_n(u)^2 du \leq C'_\Delta n^{-2} = o(n^{-1}).$$

The root- n claim then follows from proposition 2.11. The same smoothness package is consistent with the classical Bahadur theory of Bahadur [1966] and van der Vaart [1998, Chapter 21], but here the explicit deformation bounds make the integrated remainder quantitative.

Remark 2.13 (Weak rowwise dependence). The proof of proposition 2.11 uses independence within a row only through the $O(n^{-1})$ variance bound for the empirical cdf term and through item (iv). Consequently, the same root- n conclusion persists under any short-range dependent rowwise model for which the empirical cdf still has L_2 fluctuation of order $n^{-1/2}$ and the integrated Bahadur remainder remains $o(n^{-1})$. Short-range mixing conditions are one natural route to such a package, but no general dependent-row theorem is proved here. The smooth deformation corollary above is the verified instance used here; broader mixing-based verification is not developed in this paper.

Empirical diagnostics for the smooth deformation class and the fixed-support triangular-array model appear in appendix B.

2.6 Structural Lower Bound

The lower bound shows that, in the root- n finite-sample regime $a = 1/2$, the horizon is imposed by the drift geometry itself.

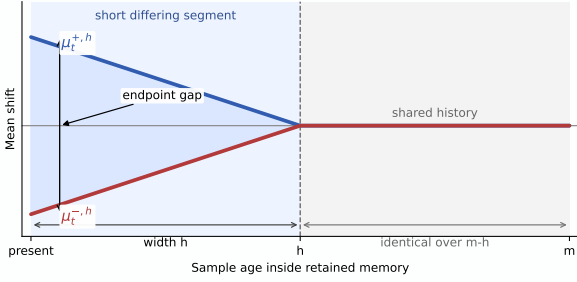


Figure 2: Least-favorable Gaussian path pair for the root- n horizon lower bound. The two paths share most of the retained history and differ only on a recent segment of width h , yet remain separated at the present. That recent segment already produces the same horizon exponent as the upper law in the root- n regime.

Proposition 2.14 (Structural lower bound in the root- n regime). *Fix $H \in (0, 1]$ and work on the Gaussian location subclass with known variance σ^2 and path pair*

$$\mu_{t-j}^{\pm,h} = \pm\beta(h-j)_+^H, \quad j = 0, 1, \dots, h,$$

where $\beta \leq \zeta$ enforces the roughness budget. Let $\mathcal{G}_{H,\zeta}^{\text{Gauss}}$ denote the corresponding Gaussian location subclass of target paths, and suppose the retained observations are independent with laws $N(\mu_{t-j}^{\pm,h}, \sigma^2)$. Then there exists a constant $c_H > 0$ depending only on H such that the endpoint tracking risk over this subclass satisfies

$$\inf_{\hat{P}} \sup_{P \in \mathcal{G}_{H,\zeta}^{\text{Gauss}}} \mathbb{E} W_2(\hat{P}_t, P_t) \geq c_H \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

Consequently, in the root- n finite-sample regime $a = 1/2$, the horizon scale

$$h^*(H) \asymp (\sigma/\zeta)^{2/(2H+1)}$$

is structural even on this restricted subclass.

Proof. Compare two hypotheses generated by the path pair $\mu_{t-j}^{+,h}$ and $\mu_{t-j}^{-,h}$. At the present time, the endpoint separation is of order βh^H . On the Gaussian location subclass, this is also the W_2 separation. The Kullback–Leibler divergence between the two hypotheses is proportional to

$$\beta^2 \sum_{r=1}^h r^{2H} / \sigma^2.$$

Choosing

$$\beta = \min \left\{ \zeta, \frac{\sigma}{2\sqrt{\sum_{r=1}^h r^{2H}}} \right\}$$

keeps the testing problem in the Le Cam regime while respecting the roughness budget. The resulting two-point lower bound is of order βh^H . Balancing the active roughness constraint against the testing constraint gives $h^*(H) \asymp (\sigma/\zeta)^{2/(2H+1)}$ and therefore the displayed lower

law. Appendix appendix A gives the exact Gaussian two-point calculation used here.

For the abstract drifting class, this lower bound is subclass based rather than class-tight, but it is enough to show that the horizon is a structural feature of the problem. The Gaussian location theorem below closes the corresponding benchmark model.

The proposition above is the lower-bound fact needed for the main horizon law. The same Gaussian construction also exposes the constant-level geometry. The following proposition gives the sharp Gaussian constants available within that proof scheme.

Appendix appendix A records the sharpened Gaussian constant geometry for the ramp and endpoint-minimal profiles.

The same lower-bound geometry also yields a full class-level benchmark once the model is specialized to Gaussian location tracking. In that setting, the deterministic Hölder envelope, the finite-memory estimator, and the lower-bound construction all live in the same one-dimensional parameterization, so this model admits a direct minimax rate statement.

Proof. For the upper bound, estimate the present mean by the uniform-window average

$$\hat{\mu}_t^{(n)} := \frac{1}{n} \sum_{j=0}^{n-1} Y_{t-j}$$

and set $\hat{P}_t^{(n)} = N(\hat{\mu}_t^{(n)}, \sigma^2)$. Because equal-variance Gaussian laws satisfy

$$W_2(N(m_1, \sigma^2), N(m_2, \sigma^2)) = |m_1 - m_2|,$$

the risk reduces to mean estimation error. Since

$$\hat{\mu}_t^{(n)} - \frac{1}{n} \sum_{j=0}^{n-1} \mu_{t-j} \sim N\left(0, \frac{\sigma^2}{n}\right),$$

the noise term is $\mathbb{E}|N(0, \sigma^2/n)| = \sigma\sqrt{2/(\pi n)}$, while the deterministic bias is bounded by

$$\frac{\zeta}{n} \sum_{j=0}^{n-1} j^H,$$

so

$$\sup_{\mu \in \mathfrak{M}_H(\zeta)} \mathbb{E} W_2(\hat{P}_t^{(n)}, P_t) \leq \sigma\sqrt{\frac{2}{\pi n}} + \frac{\zeta}{n} \sum_{j=0}^{n-1} j^H.$$

Rescaling by $n = A(\sigma/\zeta)^{2/(2H+1)}$ yields a one-parameter normalized upper envelope. Differentiating its exact Gaussian-location form gives the stated root equation for y_H , and therefore the displayed asymptotic upper constant and horizon shape. In particular, this proves the stated upper rate and horizon scale.

For the lower bound, use the symmetric two-point subfamily

$$\mu_{t-j}^{\pm,h} = \pm\beta(h^H - j^H)_+, \quad j \geq 0,$$

with $\beta \leq \zeta$. This family lies in $\mathfrak{M}_H(\zeta)$ because

$$|\mu_t^{\pm,h} - \mu_{t-j}^{\pm,h}| = \beta \min(h^H, j^H) \leq \zeta j^H.$$

Moreover, for fixed endpoint value βh^H , it minimizes the Gaussian testing energy over all symmetric lag profiles allowed by the present-point Hölder envelope, since each lag coordinate is independently minimized by $\beta(h^H - j^H)_+$. The exact Gaussian two-point calculation from the lower-bound analysis in appendix A therefore applies on the full deterministic Hölder class and yields a lower bound of order

$$\sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

Minimizing over h gives the same horizon scale $(\sigma/\zeta)^{2/(2H+1)}$. Combining the two sides proves the theorem.

2.7 Projection and Sliced Transfer

The horizon law also extends beyond one dimension when the multivariate geometry reduces to a uniformly controlled family of one-dimensional transport problems. This is the natural setting for sliced or coordinate-averaged Wasserstein constructions, where the finite-sample term is built by integrating projected one-dimensional errors.

Corollary 2.15 (Finite-family projection transfer). *Let $\Pi = \{\theta_1, \dots, \theta_m\}$ be a finite family of linear maps from the ambient space to $\mathbb{R}$, and define*

$$d_\Pi(P, Q) := \left(\frac{1}{m} \sum_{r=1}^m W_2^2(P_{\theta_r}, Q_{\theta_r}) \right)^{1/2}.$$

Assume that for every $r = 1, \dots, m$:

- (i) *the projected triangular-array window satisfies the hypotheses of proposition 2.11 with constants uniform in r ;*
- (ii) *the projected drift path obeys the same Hölder staleness envelope with exponent H and amplitude at most ζ uniformly in r .*

Then there exist constants $C_{K,\Pi}, C_{S,\Pi} < \infty$ such that

$$\mathbb{E} d_\Pi(\hat{P}_t^{(n)}, P_t) \leq C_{K,\Pi} n^{-1/2} + C_{S,\Pi} \zeta n^H.$$

Consequently,

$$n_\Pi^*(H) \asymp (C_{K,\Pi}/\zeta)^{1/(H+1/2)}.$$

In particular, the same law holds for coordinate-averaged product constructions obtained by taking Π to be the coordinate projections.

Proof. For each r , proposition 2.11 gives

$$\mathbb{E} W_2^2(\hat{P}_{t,\theta_r}^{(n)}, \bar{P}_{t,\theta_r}^{(n)}) \leq C n^{-1}$$

with C uniform in r . Averaging over r yields

$$\mathbb{E} d_\Pi^2(\hat{P}_t^{(n)}, \bar{P}_t^{(n)}) \leq C n^{-1},$$

and Jensen gives the root- n bound for $\mathbb{E} d_\Pi(\hat{P}_t^{(n)}, \bar{P}_t^{(n)})$. The projected Hölder bound averages in the same way:

$$d_\Pi(\bar{P}_t^{(n)}, P_t) \leq C_{S,\Pi} \zeta n^H.$$

Applying lemma 2.1 with $d = d_\Pi$ yields the displayed envelope and horizon scale.

Corollary 2.16 (Sliced-Wasserstein transfer under uniform projected control). *Let ρ be a probability measure on a family of one-dimensional projections θ , and define*

$$SW_{2,\rho}^2(P, Q) := \int W_2^2(P_\theta, Q_\theta) \rho(d\theta),$$

where P_θ denotes the pushforward of P under projection θ . Assume that for every θ in the support of ρ :

- (i) *the projected triangular-array window satisfies the hypotheses of proposition 2.11 with constants uniform in θ ;*
- (ii) *the projected drift path obeys the same Hölder staleness envelope with exponent H and amplitude at most ζ uniformly in θ .*

Then there exist constants $C_K, C_S < \infty$ such that

$$\mathbb{E} SW_{2,\rho}(\hat{P}_t^{(n)}, P_t) \leq C_K n^{-1/2} + C_S \zeta n^H.$$

Consequently,

$$n^*(H) \asymp (C_K/\zeta)^{1/(H+1/2)}.$$

Finite projection families are included by taking ρ supported on finitely many atoms; corollary 2.15 states that case directly for multivariate tracking.

Proof. For each projection θ , proposition 2.11 gives

$$\mathbb{E} W_2^2(\hat{P}_{t,\theta}^{(n)}, \bar{P}_{t,\theta}^{(n)}) \leq C n^{-1}$$

with C uniform in θ . Integrating over ρ yields

$$\mathbb{E} SW_{2,\rho}^2(\hat{P}_t^{(n)}, \bar{P}_t^{(n)}) \leq C n^{-1},$$

and Jensen gives the root- n bound for $\mathbb{E} SW_{2,\rho}(\hat{P}_t^{(n)}, \bar{P}_t^{(n)})$. The projected Hölder bound integrates in the same way, giving

$$SW_{2,\rho}(\bar{P}_t^{(n)}, P_t) \leq C_S \zeta n^H.$$

Applying lemma 2.1 with $d = SW_{2,\rho}$ yields the stated envelope and horizon scale.

Weighted-memory summaries and the effective-sample-size comparison are recorded in appendix A.

2.8 Operational Transport Examples

The transfer corollary above captures the operational reading.

3 Horizon Inference from Lag Geometry

Structural validity is not enough for action. A data-driven information-state rule needs estimators of the drift parameters (H, ζ) together with a criterion that compares inferential resolution to the curvature of the useful-memory region. Lag geometry supplies that inference surface: present-time validity becomes observable when discrepancies across controlled lags reveal the roughness and amplitude of the underlying drift.

Write $\alpha = \log \zeta$, and let m denote the per-lag sample size used to estimate lagged transport discrepancies. For a fixed lag design $1, \dots, L$, define the lag-energy information scale

$$\mathcal{I}_{m,L}(H) := m \sum_{j=1}^L j^{2H}.$$

Assumption 3.1 (Lag-geometry observability input). For the lag model $D_j = \zeta j^H$, suppose there are estimators $(\hat{\alpha}_{m,L}, \hat{H}_{m,L})$ such that, on compact parameter sets,

$$\sqrt{\mathcal{I}_{m,L}(H)} \begin{pmatrix} \hat{\alpha}_{m,L} - \alpha \\ \hat{H}_{m,L} - H \end{pmatrix} \Rightarrow \mathcal{N}(\mathbf{0}, \Sigma_{L,H,\zeta}).$$

The power-law lag-geometry model of [Parreira \[2026b\]](#) provides one concrete source of this assumption.

The continuous optimizer of the horizon envelope can be written as the smooth map

$$n_*(\alpha, H) = \left(\frac{aC_K}{HC_S e^\alpha} \right)^{1/(a+H)},$$

$$g(\alpha, H) := \log n_*(\alpha, H).$$

Its gradient is

$$\nabla g(\alpha, H) = \begin{pmatrix} -(a+H)^{-1} \\ -[H(a+H)]^{-1} - g(\alpha, H)(a+H)^{-1} \end{pmatrix}.$$

Theorem 3.2 (Plug-in Validity Central Limit Law). *Under Assumption 3.1,*

$$\sqrt{\mathcal{I}_{m,L}(H)} (\log \hat{n}_* - \log n_*) \Rightarrow \mathcal{N}(0, \tau^2),$$

where $\hat{n}_* = n_*(\hat{\alpha}_{m,L}, \hat{H}_{m,L})$ and

$$\tau^2 = \nabla g(\alpha, H)^\top \Sigma_{L,H,\zeta} \nabla g(\alpha, H).$$

Proof. The map g is smooth on any compact set bounded away from $H = 0$ and $\zeta = 0$. Apply the multivariate delta method to $(\hat{\alpha}_{m,L}, \hat{H}_{m,L})$ under Assumption 3.1.

The useful-memory region is the set

$$\mathcal{U}_\delta = \{n : \Phi(n) \leq (1 + \delta)\Phi(n_*)\}.$$

Let $(x_-(\delta; a, H), x_+(\delta; a, H))$ be the normalized endpoints from Proposition 2.9, and define the log-radius

$$r_\delta(a, H) := \min\{-\log x_-(\delta; a, H), \log x_+(\delta; a, H)\}.$$

This quantity is the largest symmetric log-error around n_* that stays inside the useful-memory region.

Theorem 3.3 (Operational Validity Identifiability of Useful Memory). *Under Assumption 3.1,*

$$\mathbb{P}(\hat{n}_* \in \mathcal{U}_\delta) = 2\Phi_{\mathcal{N}}\left(\frac{\sqrt{\mathcal{I}_{m,L}(H)} r_\delta(a, H)}{\tau}\right) - 1 + o(1),$$

where $\Phi_{\mathcal{N}}$ is the standard normal CDF. In particular,

$$\frac{\sqrt{\mathcal{I}_{m,L}(H)} r_\delta(a, H)}{\tau(H, \zeta)} \rightarrow \infty \implies \mathbb{P}(\hat{n}_* \in \mathcal{U}_\delta) \rightarrow 1.$$

Proof. By Proposition 2.9, $\hat{n}_* \in \mathcal{U}_\delta$ if and only if $|\log \hat{n}_* - \log n_*| \leq r_\delta(a, H)$. Standardize this event with Theorem 3.2 and use the symmetry of the Gaussian limit.

The quantity

$$\mathfrak{S}_{\delta,m,L} := \frac{\sqrt{\mathcal{I}_{m,L}(H)} r_\delta(a, H)}{\tau(H, \zeta)}$$

is the operational identifiability score of the horizon. It compares lag-geometry information to the validity-curvature-imposed precision requirement of the useful-memory region.

The following algorithm summarizes the operational pipeline.

Algorithm 1 Operational validity-observability pipeline

- 1: Estimate recent lag discrepancies on lags $1, \dots, L$.
 - 2: Fit the lag model to obtain $(\hat{\alpha}_{m,L}, \hat{H}_{m,L})$ and a covariance proxy.
 - 3: Compute $\hat{n}_*$ and $\hat{\mathfrak{S}}_{\delta,m,L}$.
 - 4: **if** $\hat{\mathfrak{S}}_{\delta,m,L}$ is below the target tolerance threshold **then**
 - 5: widen to the useful-memory band or defer the update
 - 6: **else**
 - 7: deploy $\hat{n}_*$ and monitor realized tracking error
 - 8: **end if**
-

Proposition 3.4 (Deterministic stability of the horizon map). *Let $K \subset \mathbb{R} \times (0, \infty)$ be compact. Then there exists a finite constant L_K such that, for all $(\alpha_1, H_1), (\alpha_2, H_2) \in K$,*

$$|g(\alpha_1, H_1) - g(\alpha_2, H_2)| \leq L_K (|\alpha_1 - \alpha_2| + |H_1 - H_2|).$$

Consequently, if (α, H) and $(\hat{\alpha}_{m,L}, \hat{H}_{m,L})$ lie in K and

$$|\hat{\alpha}_{m,L} - \alpha| + |\hat{H}_{m,L} - H| \leq \frac{r_\delta(a, H)}{L_K},$$

then $\hat{n}_* \in \mathcal{U}_\delta$.

Proof. The gradient of g is continuous on K , so $\|\nabla g\|_1$ is bounded there. Apply the mean-value theorem to obtain the Lipschitz bound. The inclusion claim follows from $|\log \hat{n}_* - \log n_*| \leq r_\delta(a, H)$ together with the characterization of $\mathcal{U}_\delta$ from Proposition 2.9.

The informative regime lies across the transition from low to moderate identifiability. Table 2 shows representative points across that transition.

The lag-geometry pipeline is accurate on the power-law family used in the assumption, but it is not stable under arbitrary departures from that family. Targeted Monte Carlo checks on sinusoidal and piecewise departures show a small bias under the exact model and a much larger bias in both $\hat{H}$ and the induced horizon estimate when the lag law is misspecified. Table 1 summarizes that sensitivity.

scenario	H bias	$\log n^*$ bias	$\log n^*$ RMSE
correct	-0.0000	+0.0002	0.0090
sinusoidal	-0.0932	+0.1100	0.1104
piecewise	-0.2114	+0.8291	0.8292

Table 1: Misspecification probe for observability.

On the tested grid, the plug-in central limit law is also well calibrated at the scale level: the empirical standard deviation of $\log \hat{n}_*$ tracks the delta-method prediction closely, and the nominal 95% Gaussian interval coverage stays near its target across the reported regimes. The simulation suite also calibrates the regret expansion directly.

Table 2: Representative regimes across the inferable-horizon transition.

L	m	H	δ	$\mathfrak{S}$	Empirical	Gaussian
5	20	0.3	0.050	0.501	0.411	0.384
8	10	0.9	0.020	1.001	0.633	0.683
3	200	0.3	0.010	1.996	0.956	0.954
8	100	0.3	0.005	3.001	0.996	0.997

Proposition 3.5 (Deterministic and asymptotic plug-in horizon regret). *Let $u = \log(\hat{n}_*/n_*)$. Then, as $u \rightarrow 0$,*

$$\frac{\Phi(\hat{n}_*) - \Phi(n_*)}{\Phi(n_*)} = \Psi(e^u) - 1 = \frac{He^{-au} + ae^{Hu}}{a + H} - 1.$$

Consequently, for every $r > 0$,

$$|u| \leq r \implies \frac{\Phi(\hat{n}_*) - \Phi(n_*)}{\Phi(n_*)} \leq B_r(a, H),$$

where

$$B_r(a, H) := \max\{\Psi(e^r) - 1, \Psi(e^{-r}) - 1\}.$$

Moreover,

$$\frac{\Phi(\hat{n}_*) - \Phi(n_*)}{\Phi(n_*)} = \frac{aH}{2}u^2 + O(u^3).$$

Consequently,

$$\mathbb{E} \left[\frac{\Phi(\hat{n}_*) - \Phi(n_*)}{\Phi(n_*)} \right] = \frac{aH}{2} \frac{\tau^2}{\mathcal{I}_{m,L}(H)} + o(\mathcal{I}_{m,L}(H)^{-1}).$$

Proof. By Proposition 2.9,

$$\frac{\Phi(n_*e^u)}{\Phi(n_*)} = \Psi(e^u) = \frac{He^{-au} + ae^{Hu}}{a + H}.$$

The deterministic bound follows by maximizing this exact expression over $u \in [-r, r]$. Differentiate at $u = 0$. The first derivative vanishes and the second derivative equals aH , giving the quadratic expansion. The expectation bound then follows from Theorem 3.2 and the boundedness of third moments on compact parameter sets.

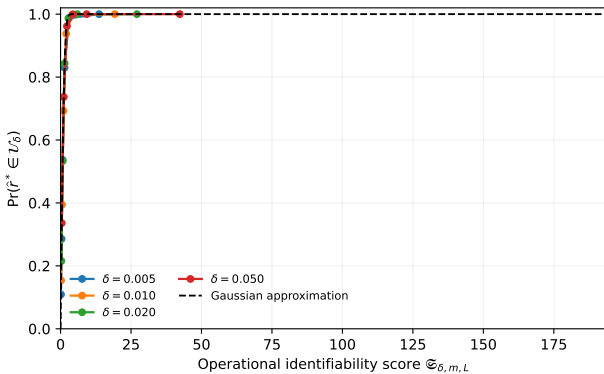


Figure 3: Operational identifiability of the useful-memory region. Binning by the operational score reveals a monotone rise in useful-region hit rate, with each δ curve tracking the Gaussian approximation from Theorem 3.3.

Structural existence and operational identifiability need not coincide. The relevant scale is the score $\mathfrak{S}_{\delta,m,L}$, which organizes the transition from low coverage to near-certain useful-memory recovery.

4 Information-State Transfer, Calibration, and Control

The horizon law is attached to an information state, not only to a raw retention window. Once a pipeline admits a finite-sample envelope and a staleness envelope, the same constitutive balance governs estimation, calibration, and control. Finite projection families and sliced geometries transfer the law under uniform projected control. Beyond those proved classes, high-dimensional practicality becomes a geometric question: temporal validity remains inferable only when the chosen state geometry preserves a usable finite-sample exponent.

Corollary 4.1 (Effective-exponent horizon transfer). *Let d_{eff} be a probability metric such that, on the drifting class of interest,*

$$\mathbb{E} d_{\text{eff}}(\hat{P}_t^{(n)}, \bar{P}_t^{(n)}) \leq C_{K,\text{eff}} n^{-a_{\text{eff}}}$$

for some $a_{\text{eff}} > 0$, and suppose the same class satisfies the staleness bound

$$d_{\text{eff}}(\bar{P}_t^{(n)}, P_t) \leq C_{S,\text{eff}} \zeta n^H.$$

Then

$$\mathbb{E} d_{\text{eff}}(\hat{P}_t^{(n)}, P_t) \leq C_{K,\text{eff}} n^{-a_{\text{eff}}} + C_{S,\text{eff}} \zeta n^H,$$

and the induced validity horizon satisfies

$$n_{\text{eff}}^*(H) \asymp (C_{K,\text{eff}}/\zeta)^{1/(a_{\text{eff}}+H)}.$$

Proof. Apply Lemma 2.1 with $d = d_{\text{eff}}$.

The relevant question is which geometry preserves a usable exponent a_{eff} . Finite projection and sliced constructions provide explicit examples with one-dimensional finite-sample control. Intrinsic-dimension and projection-robust transport can replace ambient dimension by an effective dimension, while smoothed transport can preserve parametric behavior through regularization [Weed and Bach, 2019, Paty and Cuturi, 2019, Nietert et al., 2022]. In these broader settings, the effective-exponent corollary yields a horizon law; it does not imply a new multivariate transport theorem.

A practical guide is therefore geometry-first. When the ambient dimension is low and the geometry is well resolved, raw Wasserstein remains the most direct choice. When only a small set of directions carries drift, finite projections or sliced variants are preferable because they preserve a usable exponent by discarding uninformative directions. When the geometry is noisy or the support is moderate, a calibrated regularization band is natural because it regularizes the finite-sample term without forcing a new transport theorem. In all cases, the guiding criterion is whether the chosen geometry supplies an effective exponent large enough to keep the horizon nondegenerate.

An equivalent operator view is often more informative than the geometry-first view. Let $R_t^{(n)}$ retain a state from the recent past, let U_t update that state as new observations arrive, and let G_t extract an estimate, interval, alarm statistic, or control action. The object governed by validity is the composite information state $G_t \circ U_t \circ R_t^{(n)}$, not

memory length alone. The horizon law explains when a retained state remains useful, the inference layer estimates that horizon from lag geometry, and the control layer decides whether to deploy a point estimate, a useful-memory band, or a deferred update.

Conformal calibration under drift fits this template directly. A split-conformal calibration set is a retained information state, the nonconformity quantile is a readout, and predictive coverage is the operational validity criterion. The same finite-sample versus staleness balance therefore applies to calibration-window selection. In the benchmark reported here, the calibration score has an interior optimum over window size, coverage degrades before alarm, and the empirically safe calibration band coincides with the useful-memory band. The conformal connection is therefore constitutive rather than analogical: calibration under drift is another readout of validity.

Sequential memory control closes the same object on the action side. Once memory becomes a controlled state and updates incur fixed and proportional actuation cost, the decision problem is not to maximize memory or to react as early as possible, but to remain inside the temporally valid region at acceptable cost. The resulting frontier is regime dependent. Drift roughness, sensing quality, and actuation price determine whether aggressive control, detector-triggered updates, or more conservative deployment is preferred.

Weighted-window diagnostics on uniform, triangular, and geometric tapers show the same finite-sample versus lag trade-off in operational form; see Table 13. Horizon inference begins with the plug-in horizon entering the useful-memory region and remaining there under streaming noise. Support change beyond the fixed support-growth construction and online validity control under streaming noise are not treated here at theorem level.

5 Operational Use of Validity

The validity horizon becomes operational once memory is treated as a controlled state. The same finite-sample versus staleness balance that defines the horizon also determines when an online controller should update, defer, or hold its memory scale. The resulting problem is not to maximize memory or to react as soon as possible, but to keep the information state inside its useful band at acceptable cost.

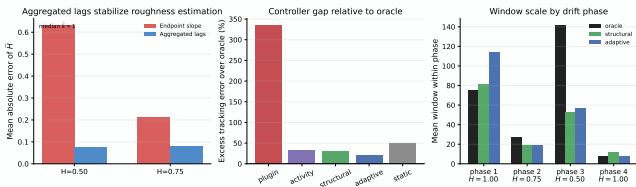


Figure 4: Online implications of the validity horizon. Aggregated lag stabilizes roughness estimation, the structural controller closes most of the gap to the oracle policy, and the preferred memory scale rotates across drift phases.

The figure above summarizes the online object directly. Aggregating lagged discrepancies improves roughness estimation, the structural controller reduces tracking error relative to the raw plug-in rule, and the preferred scale moves with the drift phase rather than remaining fixed.

The online use of temporal validity is therefore a combined inference-and-control problem.

Profile	Best static/oracle	Activity/oracle	EWMA/oracle	Kalman/oracle	Plug-in/oracle	Structural/oracle	Adaptive/oracle
Default	1.43	1.36	1.34	1.28	4.41	1.36	1.30
Smooth	1.05	1.41	1.37	1.31	3.24	1.22	1.24
Rough	1.09	1.65	1.64	1.58	2.59	1.30	1.39
Alternating	1.02	1.65	1.60	1.56	2.30	1.24	1.36
Ramp-up	1.02	1.34	1.28	1.21	2.91	1.18	1.23

Table 3: Online memory-selection baselines across five profile families, aggregated over seeds 0, 1, 2, with maximum one-step window-index jumps. Entries report mean absolute error relative to the oracle controller. The structural controller consistently improves on the raw plug-in rule, beats the adaptive EWMA baseline on 13/15 runs and the adaptive scalar Kalman baseline on 11/15 runs, and is strongest on the rough and alternating profiles.

Across the five tested profile families, the structural controller consistently improves on the raw plug-in rule and beats the adaptive EWMA and adaptive Kalman baselines on the majority of runs. The gains are strongest on the rough and alternating profiles, where the online memory state must move with drift while still resisting noise. This is the operational interpretation of the validity horizon: a memory scale is useful only while its retained information remains temporally valid, and the controller must pay cost to keep it there.

The controller evidence is therefore not a separate story from the horizon law. It is the sequential readout of the same object. Horizon inference estimates the usable band; online control decides whether to deploy it, hold it, or update it under cost.

6 Empirical Signatures

The experiments target qualitative signatures of the validity field: an interior optimum over memory scale, roughness-dependent horizon movement, a real-stream analogue, positive separation between validity loss and alarm, regime-dependent operational control, and an explicit conformal-calibration readout. The reported sweeps use repeated Monte Carlo runs and are meant to identify shape, ordering, and sign rather than to estimate constants precisely.

6.1 Interior Memory Optimum

The Gaussian horizon sweep isolates the variance–staleness trade-off directly. Small windows are variance dominated, large windows are stale, and the minimum lies at an interior horizon.

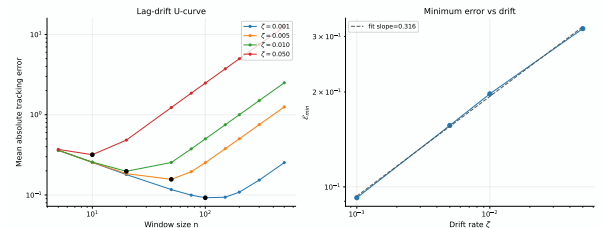


Figure 5: Interior risk optimum over memory length. Tracking error first falls with horizon and then rises once staleness overtakes the remaining variance gain. The optimum remains finite across drift strengths.

This interior optimum is the basic empirical signature of validity under drift. Under stationarity, more memory

would continue to help. Under drift, the horizon that reduces variance also ages the evidence. Around the minimizing horizon, the curve displays a flat near-optimal region, which is the empirical analogue of the useful-memory region defined from the envelope Φ .

6.2 Roughness-Dependent Horizon Scaling

The roughness-family experiment varies the roughness index directly. For each Hölder-type regime, the surrogate error obeys the same structural decomposition as the theory,

$$\mathcal{E}(n) \approx \sigma n^{-1/2} + \zeta n^H,$$

and the experiment traces the empirically optimal horizon across a sweep of ζ values.

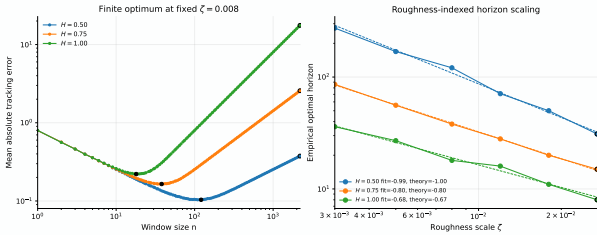


Figure 6: Roughness-dependent horizon scaling. Each path class exhibits a finite optimum, and the location of that optimum changes with H . The horizon is a function of drift regularity, not a fixed window rule.

The empirical optima separate by roughness class, matching the path-dependent horizon logic of corollary 2.5.

6.3 Real-Stream Memory Diagnostic

A similar signature appears on a real nonstationary stream. On ELEC2 [Harries, 1999], the diagnostic compares a short anchor block with the preceding memory window in W_1 , yielding a real-data analogue of the finite-memory U-curve.

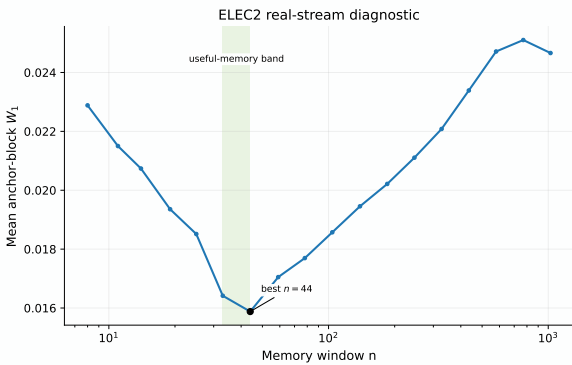


Figure 7: ELEC2 real-stream diagnostic on the `nswprice` marginal. The mean anchor-block W_1 error exhibits a finite interior optimum and a near-optimal useful-memory band, matching the validity signature on a real stream.

The optimum occurs at a finite memory scale with a short near-optimal band. The same retention pattern therefore appears on a real stream: very short windows are too noisy, while very long windows become stale relative to the present block. On the robustness grid, the `nswprice` marginal retains an interior optimum in all tested configurations, and the `nswdemand` marginal shows the same inte-

rior pattern with more movement in the optimizing scale; see Table 10.

6.4 Validity Loss Before Detection

Temporal validity and changepoint evidence are different clocks. To make that separation explicit, consider a stream whose local drift rate ramps upward over time. The horizon n_t^* contracts as the ramp steepens, while a long operating horizon remains fixed. Define

$$\Delta_{\text{val-det}} := t_{\text{detect}} - t_{\text{valid}},$$

where t_{valid} is the first time the operating horizon persistently exceeds the validity horizon, and t_{detect} is the first detector alarm that follows.

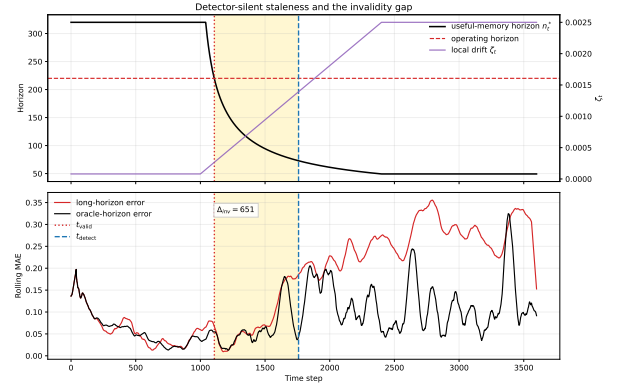


Figure 8: Temporal validity can fail before detector alarm. As drift intensifies, the validity horizon contracts until the operating horizon becomes stale at t_{valid} ; the detector responds only later at t_{detect} , leaving a strictly positive validity-detection lag. The lower panel shows excess tracking error accumulating before the alarm appears.

Across the reported runs, the measured lag remains positive. Retained evidence can already be invalid for the present while detector statistics remain below alarm threshold.

The positive lag is not a knife-edge feature of one tuning choice. A compact false-alarm-calibrated frontier over $H \in \{0.5, 0.75, 1.0\}$, operating windows 160 and 240, roughness levels 0.01 and 0.024, and detector families ADWIN, Page-Hinkley, KSWIN, and CUSUM shows the same separation on the tested observation-based detector families. Observation-based ADWIN and Page-Hinkley retain unit detection rate and strictly positive mean gaps on the successful grid points, while KSWIN preserves the sign of the gap when it detects but with substantially lower detection rates. Residualized detector inputs are not uniformly better on this family. The calibrated sign result therefore holds for observation-based detector families on the tested compact grid, not as a detector-uniform delay law; see Table 11.

6.5 Conformal Calibration under Drift

The same geometry appears when retained information is used for predictive coverage rather than for direct distribution tracking. In the split-conformal benchmark, the calibration window acts as a finite-memory state, the non-conformity quantile is the readout, and the score combines

interval width with coverage loss. Small calibration windows are variance dominated, while large calibration windows become stale under drift, so the score has an interior optimum over window size.

The benchmark also separates the calibration-validity clock from the detection clock. Coverage loss occurs before alarm on the tested drifted residual family, leaving a positive coverage-before-alarm gap. Around the score-minimizing window, the empirically safe calibration band coincides with the useful-memory band. Conformal prediction under drift therefore fits the same validity law: retained calibration is useful only up to a finite horizon.

6.6 Empirical Summary

The experiments support a structural validity profile with a near-optimal useful-memory region, roughness-dependent horizon scaling, a corresponding pattern on the ELEC2 stream, a measurable lag between validity loss and detector alarm, and a conformal calibration window with the same horizon geometry. The online adaptation section summarizes the controller and phase-map evidence; it shows that the structural controller keeps mean error close to the oracle while the preferred scale rotates with drift phase.

7 Conclusion

Validity under drift is constrained by the validity of the information state used at the present time. Retention reduces finite-sample noise and ages the retained state simultaneously, so validity has a finite horizon.

In the deterministic H^* -older–Wasserstein constitutive model, the horizon law

$$\mathbb{E} d(\hat{P}_t^{(n)}, P_t) \leq C_K n^{-a} + C_S \zeta n^H,$$

and its optimizer

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}$$

summarize that balance. This law induces a useful-memory region around the optimal scale.

A structural Gaussian lower bound and a Gaussian location benchmark identify the horizon as a property of drift geometry rather than of a particular estimator. A tractable one-dimensional proof anchor and projection-based transfer establish the same mechanism beyond the benchmark class.

The empirical signatures align with the theory: U-curves, roughness-dependent horizon scaling, a real-stream finite-memory pattern, a positive lag between validity loss and detector alarm, a regime-dependent policy frontier under actuation cost, and a conformal calibration window with the same finite-memory geometry. Horizon inference from lag geometry captures the observability side: validity can be structurally defined while remaining hard to estimate from data alone.

Inferential persistence therefore has a measurable profile with a critical scale under flow. The central object is the validity of the information state itself. Memory choice, calibration-window choice, and sequential control are all operational readouts of that same object.

8 Limitations and Open Problems

The proved results cover the validity horizon, its useful-memory geometry, the one-dimensional proof anchor, the Gaussian lower and benchmark results, and horizon-inference guarantees.

The remaining open problems are the following.

- (1) **Class-tight lower theory beyond the Gaussian benchmark.** The abstract lower proposition establishes that the validity horizon is structural by means of a Gaussian two-point subclass, and the Gaussian location benchmark closes the corresponding minimax rate on the deterministic Hölder class. What remains open is class-tight lower theory for broader drifting classes beyond that benchmark model.
- (2) **Transfer beyond the proved projected range.** Finite-family projection transfer is proved under uniform projected control, and the effective-exponent corollary shows how any geometry with a proved finite-sample exponent inherits a horizon scale. What remains unproved is an analogous theorem for richer dominated families and for raw multivariate Wasserstein, where the effective exponent can become too small for practical geometry selection.
- (3) **Operator-level transfer beyond the proved classes.** The current theory closes the law for benchmark tracking models, finite-family projection transfer, sliced transfer under uniform projected control, and the embedded transport examples developed here. A general operator-level theorem for richer information states, including adaptive calibration and broader multivariate transport regimes, remains open.
- (4) **Online validity control under streaming noise.** Horizon inference closes plug-in asymptotics, identifiability, deterministic stability, and quadratic regret under the lag-geometry input model. The open operational problem is to obtain stable online estimation and time-uniform memory localization guarantees under streaming noise. The experiments target structural signatures rather than detector-uniform delay laws or competitive end-to-end performance, so sequential validity versus detection remains open at the level of general theory.
- (5) **Lag-model misspecification in observability.** The observability pipeline assumes a power-law lag profile. In targeted Monte Carlo checks, sinusoidal and piecewise departures from that law produced small bias under the exact model but much larger bias in both $\hat{H}$ and the induced horizon estimate when the law was misspecified. A robust theory for horizon inference under such departures remains open.
- (6) **Conformal calibration under drift at theorem level.** The split-conformal benchmark exhibits the same validity geometry as distribution tracking, but the current conformal connection is empirical. A theorem-level transfer from information-state validity to time-varying coverage, set width, and calibration-window selection under drift remains to be developed.
- (7) **Validity surface across flow dimensions.** The main law is stated for a one-parameter flow class, but

the same algebra suggests a surface $V(\phi, \tau_d)$ indexed by dimensions such as time, space, structure, and operation. A general theorem that makes this multi-axis surface explicit, with validity as one slice, remains conjectural.

9 Related Work

Drifted optimization and regret. Dynamic-regret and variation-budget analyses show that under drift, forgetting scales and window lengths must adapt to how quickly the target moves [Besbes et al., 2015, Cheung et al., 2019, Yuan and Lamperski, 2020, Mazzetto and Upfal, 2023, Wang et al., 2024]. The validity functional asks how long retained evidence remains valid before age becomes the dominant source of tracking error.

Singular learning theory provides static equations of state for inference in nonregular models [Watanabe, 2009]. The validity functional plays the dynamic role: it couples finite-sample geometry and drift regularity into a single persistence law under nonstationarity.

The same structural trade-off also appears in drifting density estimation. Mazzetto and Upfal [2023] derive min-max rates by optimizing over recent-memory windows under bounded drift, which places the present horizon law in direct contact with finite-memory risk optimization under nonstationarity.

Adaptive windows and detection. Adaptive-window and concept-drift methods study when accumulated evidence is sufficient to react [Bifet and Gavaldà, 2007, Gama et al., 2004, 2014]. Those works provide the reaction timescale. The horizon law adds the validity timescale: retained evidence can cease to represent the present before change statistics cross an alarm threshold. The same separation appears in calibration under drift, where a retained calibration window can lose present-time coverage before any detector signals that the state is stale. Adaptive-window conformal methods make a similar calibration-window choice from coverage proxies; the validity surface makes that choice geometric rather than proxy-only.

Sequential nonparametric testing. Sequential two-sample testing and kernel change detection quantify the evidence needed to reject stability under time-uniform false-alarm control. These procedures operate on a detection timescale. The validity horizon identifies the tracking timescale on which retained evidence ceases to represent the present, so the two clocks need not agree. Exponentially weighted MMD with implicit forgetting scales while still calibrating a detection threshold [Kalinke et al., 2025].

The closest stability-only perspective is the nonparametric stability principle of Huang and Wang [2023], which organizes learning under nonstationarity without an explicit constitutive law in the exponents (a, H) . The validity functional makes that geometry explicit through a closed-form persistence surface.

Resource-aware scheduling under drift, including RC-CDA [Piaseczny et al., 2025], and memory hedging across multiple forgetting scales, such as MELO-style horizon grids, address a related operational question: when should adaptation consume budget on an updated memory scale

rather than on a fixed one? The validity horizon plays the complementary role of a constitutive prior for that choice. It predicts the memory scale that a budget-aware controller should target once the effective roughness parameters are known. Variation budgets and path-length quantities are alternate summaries of the same nonstationarity resource, and translating them to (a, H) remains a natural bridge problem.

Filtering and local smoothing. Classical adaptive filtering and state-space tracking tune forgetting factors, gain schedules, or process-noise levels to balance smoothing against responsiveness [Kalman, 1960, Jazwinski, 1970]. Locally stationary smoothing and bandwidth selection, together with dynamic-regret analyses for time-varying statistical estimation, solve a closely related trade-off for evolving regression or spectral targets [Dahlhaus, 1997, Vogt, 2012]. The validity horizon places that trade-off inside a persistence field for retained information, where the finite-sample exponent depends on the chosen probability geometry or readout functional and the resulting balance defines a measurable validity geometry.

Transport geometry. Wasserstein distance and related transport geometries provide the finite-sample geometries that enter the horizon law [Fournier and Guillin, 2015, Boissard and Le Gouic, 2014, Weed and Bach, 2019, Genevay et al., 2019, Mena and Niles-Weed, 2019, del Barrio et al., 2023, Goldfeld et al., 2024]. Sliced and projection-robust variants show that geometry choice can change the finite-sample exponent, while intrinsic dimension and regularization calibration determine which regime is available [Kolouri et al., 2018, Deshpande et al., 2019, Paty and Cuturi, 2019, Lin et al., 2020, Nadjahi et al., 2020, Nietert et al., 2022, Genans and Wintenberger, 2026, Facco et al., 2017]. Those geometries sit inside a common validity law. In that law, dimensionality matters only through the effective finite-sample exponent: raw multivariate Wasserstein can force slow rates, while projection, intrinsic-dimension, or regularization mechanisms can preserve a nondegenerate horizon when they supply a usable exponent. The proved results cover finite-projection and sliced transfer together with the operational classes; the broader high-dimensional message is an effective-exponent interpretation, not a claim that any single geometry is canonical.

Age and drift. Wasserstein nonstationarity measures and Age-of-Information formulations make age into an operational variable [Jiang et al., 2020, Keehan et al., 2025, Amini and Vinas, 2026, Kaul et al., 2012, Yates et al., 2021]. The validity horizon gives that variable a distributional interpretation: age is costly because it accumulates geometric mismatch with the present.

Recent non-equilibrium accounts of continual learning describe stability through metastable escape across energy barriers [Kim, 2026]. The validity horizon is the inference-side analogue: it identifies the critical persistence scale at which retained structure ceases to track the present.

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A Proof Details

This appendix gives the detailed derivations for structural results.

A.1 Gaussian lower-bound constants

Let

$$S_h := \sum_{r=1}^h r^{2H} \quad \text{and} \quad G_h := \sum_{r=1}^h (h^H - (h-r)^H)^2.$$

For the Gaussian two-point model with equal priors and common covariance $\sigma^2 I_h$, the Bayes classification error between means $\pm m$ is $\Phi(-\|m\|/\sigma)$. Therefore the ramp and endpoint-minimal payoffs are exactly

$$L_h^{\text{ramp}}(\sigma, \zeta) = \zeta h^H \Phi\left(-\frac{\zeta}{\sigma} \sqrt{S_h}\right)$$

and

$$L_h^{\text{min}}(\sigma, \zeta) = \zeta h^H \Phi\left(-\frac{\zeta}{\sigma} \sqrt{G_h}\right).$$

Proposition A.1 (Gaussian lower-bound constants). *Fix $H \in (0, 1]$ and set $p_H := 2H/(2H+1)$. Let $x_H > 0$ be the unique maximizer of $x^{p_H} \Phi(-x)$, equivalently the unique solution of*

$$p_H \Phi(-x_H) = x_H \varphi(x_H).$$

Then, as $\sigma/\zeta \rightarrow \infty$,

$$\sup_{h \geq 1} L_h^{\text{ramp}}(\sigma, \zeta) \sim C_H^{\text{ramp}} \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)},$$

with

$$C_H^{\text{ramp}} = (2H+1)^{H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

Among all endpoint-saturating Hölder profiles, the profile

$$g_r^{\text{min}} = h^H - (h-r)^H$$

minimizes the testing energy and has asymptotic energy constant

$$I_H = \int_0^1 (1 - (1-u)^H)^2 du = \frac{2H^2}{(H+1)(2H+1)}.$$

Moreover,

$$\sup_{h \geq 1} L_h^{\text{min}}(\sigma, \zeta) \sim C_H^{\text{min}} \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)},$$

with

$$C_H^{\text{min}} = I_H^{-H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H),$$

and for $H < 1$ the improvement factor is

$$\frac{C_H^{\text{min}}}{C_H^{\text{ramp}}} = \left(\frac{H+1}{2H^2}\right)^{H/(2H+1)} > 1.$$

Proof of proposition A.1. Fix $H \in (0, 1]$ and write $p_H = 2H/(2H+1)$.

For the ramp profile,

$$S_h \sim \frac{h^{2H+1}}{2H+1}.$$

Set

$$h = A \left(\frac{\sigma}{\zeta}\right)^{2/(2H+1)}$$

with $A > 0$ fixed. Then

$$\frac{\zeta}{\sigma} \sqrt{S_h} \rightarrow \frac{A^{H+1/2}}{\sqrt{2H+1}}.$$

At the same time,

$$\zeta h^H = A^H \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

Hence

$$\frac{L_h^{\text{ramp}}(\sigma, \zeta)}{\sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}} \rightarrow A^H \Phi\left(-\frac{A^{H+1/2}}{\sqrt{2H+1}}\right).$$

Define

$$x = \frac{A^{H+1/2}}{\sqrt{2H+1}}.$$

Then

$$A^H = (2H+1)^{H/(2H+1)} x^{p_H},$$

so the normalized limiting payoff becomes

$$(2H+1)^{H/(2H+1)} x^{p_H} \Phi(-x).$$

Its logarithmic derivative is

$$\frac{p_H}{x} - \frac{\varphi(x)}{\Phi(-x)}.$$

Since the inverse Mills ratio $\varphi(x)/\Phi(-x)$ is strictly increasing on $(0, \infty)$, this objective has a unique maximizer $x_H > 0$, characterized by

$$p_H \Phi(-x_H) = x_H \varphi(x_H).$$

This proves

$$\sup_{h \geq 1} L_h^{\text{ramp}}(\sigma, \zeta) \sim C_H^{\text{ramp}} \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}$$

with

$$C_H^{\text{ramp}} = (2H+1)^{H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

For the endpoint-minimal profile, the Hölder constraint and endpoint condition imply

$$g_r \geq g_h - (h-r)^H = h^H - (h-r)^H$$

for every feasible endpoint-saturating profile g . Therefore the profile

$$g_r^{\text{min}} = h^H - (h-r)^H$$

minimizes testing energy pointwise and hence minimizes $\sum_{r=1}^h g_r^2$. Its rescaled energy satisfies the Riemann-sum limit. Thus

$$h^{-(2H+1)} G_h \rightarrow I_H,$$

where

$$I_H = \int_0^1 (1 - (1 - u)^H)^2 du = \frac{2H^2}{(H+1)(2H+1)}.$$

With the same scaling

$$h = A \left(\frac{\sigma}{\zeta} \right)^{2/(2H+1)},$$

one has

$$\frac{\zeta}{\sigma} \sqrt{G_h} \rightarrow A^{H+1/2} \sqrt{I_H}.$$

Setting $x = A^{H+1/2} \sqrt{I_H}$ gives

$$A^H = I_H^{-H/(2H+1)} x^{p_H},$$

so the normalized limiting payoff becomes

$$I_H^{-H/(2H+1)} x^{p_H} \Phi(-x).$$

The same maximizer x_H therefore yields

$$\sup_{h \geq 1} L_h^{\min}(\sigma, \zeta) \sim C_H^{\min} \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}$$

with

$$C_H^{\min} = I_H^{-H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

Finally,

$$\frac{C_H^{\min}}{C_H^{\text{ramp}}} = \left(\frac{I_H^1}{2H+1} \right)^{H/(2H+1)} = \left(\frac{H+1}{2H^2} \right)^{H/(2H+1)}.$$

This factor is strictly larger than 1 for $H < 1$ and equal to 1 at $H = 1$.

A.2 Two-point lower bound used in proposition 2.14

The structural lower bound above is the exponent-level consequence of the exact Gaussian two-point calculation. For the ramp pair

$$\mu_{t-j}^{\pm, h} = \pm \zeta (h - j)_+^H,$$

the present-time separation is $2\zeta h^H$, so any estimator incurs risk at least of order $L_h^{\text{ramp}}(\sigma, \zeta)$. Minimizing over h and using the asymptotic just proved gives the lower scale

$$\sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}$$

and therefore the horizon exponent $(2H+1)^{-1}$ for the root- n regime.

A.3 Weighted-memory summaries

For a finite-memory estimator with weights $w_0, \dots, w_{n-1}$, define the effective sample size

$$n_{\text{eff}}(w) := \left(\sum_{j=0}^{n-1} w_j^2 \right)^{-1}$$

and the weighted lag moment

$$M_H(w) := \sum_{j=0}^{n-1} j^H w_j.$$

Any weighted horizon law of the form

$$\mathbb{E} d\left(\hat{P}_t^{(w)}, P_t\right) \leq C_K n_{\text{eff}}(w)^{-a} + C_S \zeta M_H(w)$$

balances the finite-sample and staleness terms through these two summaries.

Proposition A.2 (Uniform weights maximize effective sample size). *Let $w_0, \dots, w_{n-1}$ be nonnegative weights with $\sum_{j=0}^{n-1} w_j = 1$. Then*

$$n_{\text{eff}}(w) \leq n,$$

with equality if and only if $w_j = 1/n$ for all j . Moreover, if the drift profile is an exact linear ramp in the sense that the lag- j contribution is proportional to j , then the weighted staleness equals the effective lag.

Proof. By Cauchy-Schwarz,

$$1 = \left(\sum_{j=0}^{n-1} w_j \right)^2 \leq n \sum_{j=0}^{n-1} w_j^2,$$

so $n_{\text{eff}}(w) = 1/\sum_j w_j^2 \leq n$, with equality exactly when the weights are all equal. On an exact linear ramp, averaging the lag contribution with weights w_j recovers the weighted mean lag.

B Finite-Sample and Operational Diagnostics

This appendix gives finite-sample diagnostics for the proved finite-sample results and the operational implications.

B.1 Finite-Sample Geometry Diagnostic

For the finite-sample term, we estimate empirical W_2 between two i.i.d. samples from the same distribution using exact assignment in low and moderate dimension, then fit a log-log slope in n . We also probe the fixed-span one-dimensional triangular window directly by comparing a triangular-array sample, with one draw from each drifted component, against an i.i.d. sample from the same window mixture. In that diagnostic the within-window span ζn^H is held fixed. Finally, we probe the Sinkhorn diagnostic at a few fixed values of ε .

Table 4: Empirical scaling diagnostic for the finite-sample term. The bounded-support i.i.d. W_2 experiment is close to the expected $n^{-1/2}$ rate in low dimension, the corresponding triangular-array diagnostic compares one sample from each drifted component against an i.i.d. sample from the window mixture, and the Sinkhorn diagnostic is reported at fixed ε . These rows are numerical support for the operational finite-sample discussion rather than theorem statements.

Setting	Estimated slope	Comment
$W_2, d = 1$	-0.48	close to $-1/2$
$W_2, d = 2$	-0.41	slower decay
$W_2, d = 4$	-0.25	slower decay
$W_2, d = 5$	-0.24	slower decay
W_2 , i.i.d. mixture, $d = 1, H = 0.5$	-0.50	fixed span $\zeta n^H = 0.25$
W_2 , triangular, $d = 1, H = 0.5$	-0.45	fixed span $\zeta n^H = 0.25$
W_2 , i.i.d. mixture, $d = 1, H = 1.0$	-0.47	fixed span $\zeta n^H = 0.25$
W_2 , triangular, $d = 1, H = 1.0$	-0.41	fixed span $\zeta n^H = 0.25$
Sinkhorn $\varepsilon = 0.50$	-0.21	stability constant changes
Sinkhorn $\varepsilon = 0.10$	-0.53	stability constant changes
Sinkhorn $\varepsilon = 0.05$	-0.67	stability constant changes

In low dimension, the raw i.i.d. Wasserstein term is compatible with a root- n rate, whereas higher-dimensional raw Wasserstein slows down. In the one-dimensional fixed-span triangular-array diagnostic, the recovered slopes stay near the corresponding i.i.d. mixture baseline, which is consistent with the proof-model and design-effect reading used in the 1D proof-model proposition.

B.2 Bahadur Remainder Diagnostics

Two deformation families make the proof-model remainder assumption concrete. The first is a smooth fixed-support monotone warp class $g_\delta(x) = x + \delta x(1 - x)$ with $|\delta| \leq \Delta < 1$. On the tested spans 0.2 and 0.4, the integrated remainder decays at rates between 1.40 and 1.54, and the scaled quantity $n \int_0^1 R_n(u)^2 du$ still decays at rates between 0.40 and 0.54. The second is the fixed-span triangular-array uniform-kernel model used in the proof line. There the integrated residual rate is about 1.43 for the triangular sample and 1.49 for the i.i.d. mixture benchmark, while the residual-to-MSE ratio drops from about 0.38 at $n = 25$ to about 0.11 at $n = 400$.

These diagnostics locate the role of drift span explicitly. Within the tested fixed-span range, the remainder stays lower-order and the empirical rate remains well above the n^{-1} threshold needed for item (iv). Rougher kernels and growing-span experiments remain less regular, which is why the analysis keeps the general triangular-array verification problem open.

B.3 Extended and Operational Regimes

The extended rows compare triangular and i.i.d. mixture rates on embedded low-intrinsic-dimensional support across span. The operational rows compare fixed- ε Sinkhorn rates across regularization values on embedded low-intrinsic-dimensional support in ambient dimension eight.

Table 5: Representative rows for the extended and operational finite-sample regimes. The extended rows compare triangular and i.i.d. mixture rates across span, while the operational rows compare fixed- ε Sinkhorn across ε on embedded low-intrinsic-dimensional support.

Regime	Setting	Parameter	Rate exponent α
extended	triangular ambient d=8, intrinsic k=1, span=0.10	0.10	0.4652
extended	iid mixture ambient d=8, intrinsic k=1, span=0.10	0.10	0.5722
extended	triangular ambient d=8, intrinsic k=1, span=0.25	0.25	0.4737
extended	iid mixture ambient d=8, intrinsic k=1, span=0.25	0.25	0.5588
extended	triangular ambient d=8, intrinsic k=1, span=0.50	0.50	0.4792
extended	iid mixture ambient d=8, intrinsic k=1, span=0.50	0.50	0.5425
operational	Sinkhorn iid mixture ambient d=8, intrinsic k=1, eps=0.50	0.50	0.5599
operational	Sinkhorn triangular ambient d=8, intrinsic k=1, eps=0.50	0.50	0.5706
operational	Sinkhorn iid mixture ambient d=8, intrinsic k=1, eps=0.20	0.20	0.5466
operational	Sinkhorn triangular ambient d=8, intrinsic k=1, eps=0.20	0.20	0.6150
operational	Sinkhorn iid mixture ambient d=8, intrinsic k=1, eps=0.10	0.10	0.4646
operational	Sinkhorn triangular ambient d=8, intrinsic k=1, eps=0.10	0.10	0.5060
operational	Sinkhorn iid mixture ambient d=8, intrinsic k=1, eps=0.05	0.05	0.4339
operational	Sinkhorn triangular ambient d=8, intrinsic k=1, eps=0.05	0.05	0.5124

The extended-regime rows indicate that low intrinsic dimension preserves a rate near the root- n regime without a dramatic triangular-versus-benchmark gap. The operational-regime rows record empirical evidence relevant to the fixed- ε Sinkhorn setting in ???: together with exact support-complexity inheritance and the dual threshold $\alpha > k/2$, they align with the embedded bandwise statement. The bounded-cardinality finite-support theorem closes one embedded subclass; the remaining question is support growth and support change.

Two complementary diagnostics sharpen the same boundary. First, direct probes of the centered Sinkhorn scaling Jacobian on the embedded fixed-span model show that bands approaching zero are the numerically unstable region, whereas on a moderate band such as $[0.2, 0.8]$ the cross-coupling block remains contractive and the self-coupling inverse norms appear to plateau with sample size. Second, on the supported synthetic grid, intrinsic-dimension estimates predict the observed stable-band cut-off better than ambient dimension. These diagnostics iden-

tify a useful regime statistic and isolate the band where the instability is concentrated. They probe a larger support-growth regime than the bounded-cardinality theorem.

The calibrated summary separates three ingredients on the same embedded grid: the empirically stable band, the certified band obtained from the structural certificate, and the self-coupling proxy on the calibrated pairs where that proxy is currently informative. On $(8, 2)$ and $(12, 2)$, the empirical and certified bands both reach 0.5, and the two self-coupling blocks keep positive spectral gap, worst spectral radius below 0.97, and largest-sample mean inverse norm below 3.0.

Table 6: Moderate-band Sinkhorn summary on the calibrated embedded grid. The table reports the empirical stable band, the certified band, the minimum recovered finite-sample rate exponents inside the certified band, and whether both self-coupling blocks satisfy the spectral-radius and inverse-norm proxy thresholds.

Pair	Emp. $\varepsilon_{\max}$	Thm.-ready $\varepsilon_{\max}$	Min iid α	Min tri. α	Self
(8, 1)	0.5	0.5	0.488	0.458	–
(8, 2)	0.5	0.5	0.455	0.419	yes
(12, 1)	0.2	0.2	0.501	0.474	–
(12, 2)	0.5	0.5	0.428	0.493	yes

Table 7: Calibrated moderate-band Sinkhorn certificates on the embedded fixed-span grid. The table reports the largest regularization value for which the certificate returns a certified band on the tested grid.

Ambient / intrinsic	α	Max certified ε
(8, 1)	2.0	0.5
(8, 2)	2.0	0.5
(12, 1)	2.0	0.2
(12, 2)	2.0	0.5

On the calibrated discrete $k = 2$ support grid, the support-growth diagnostics sharpen that picture further. The centered self-coupling operator for the squared block keeps radius well below one across the moderate band, the exact-target null Sinkhorn slopes are close to the n^{-1} remainder law, and the finite-support root- n empirical fluctuations remain bounded on the tested range. These diagnostics support the corrected S^2 self-coupling route and the finite-dimensional influence argument on the calibrated model. The analytic route is a finite-state minorization on the compact support rectangle: the kernel is uniformly bounded below on the calibrated band, so the centered block is a strict contraction and the inverse bound follows from a Doeblin-type decomposition. The support-growth probe points in the same direction: the ratio between exact-target Sinkhorn cost and squared ℓ_2 weight deviation stays uniformly small on the reported grid, with mean values below 0.084 and worst pointwise values below 0.24. Together with the exact $3/(4n)$ calibrated ℓ_2 fluctuation identity and the Hessian criterion in the main text, this organizes the support-growth family around a covariance-weighted trace bound and a local block-curvature bound for the null Sinkhorn map. A finite-difference basis probe supports that organization numerically: across the reported grid, local blockwise curvatures stay below about 0.24 and the stiffer collective directions scale approximately like $c(\varepsilon)n$ with reduced-channel coefficient between about 0.04 and 0.13. The local-basis trace proxy grows linearly in n at

about 0.34 per block, while the covariance-weighted proxy stays around 0.002 to 0.004.

Table 8: Calibrated support-growth Sinkhorn diagnostics on the discrete $k = 2$ support grid with span 0.25 and moderate band $\mathcal{B} = [0.1, 0.5]$. The table reports the worst centered self-coupling radius for the squared operator, the corresponding inverse-norm bound, the ranges of the exact-target null slopes for triangular and i.i.d. samples, and the worst root- n support-deviation ratio across the band.

Pair	Worst $\rho(S_z^2)$	Worst $\ (I - S_z^2)^{-1}\ $	Tri slope range	IID slope range	Worst root- n ratio
(8, 2)	0.116	1.132	$[-1.001, -0.992]$	$[-0.880, -0.878]$	2.582
(12, 2)	0.116	1.132	$[-0.878, -0.826]$	$[-0.843, -0.778]$	2.840

B.4 Smooth Multivariate Transfer Diagnostics

The multivariate transfer diagnostic uses a drifting three-dimensional mean vector and a smooth log-sum-exp functional. The finite-sample error of the window estimator, the drift-induced bias of the window target, and the linearization residual are tracked separately across window lengths. The target behavior is the same as in the abstract multivariate transfer lemma: the finite-sample piece decreases with window length, the drift piece increases, and the linearization residual remains lower order than the finite-sample term on the tested windows.

Window	Tot.	Fin.	Drift	Resid.
8	0.2009	0.1988	0.0165	0.0239
16	0.1513	0.1542	0.0355	0.0162
32	0.1373	0.1363	0.0735	0.0066
64	0.1492	0.0807	0.1503	0.0051

Table 9: Smooth multivariate transfer diagnostic under triangular-array drift. The finite-sample term decreases with window length, the drift term increases, the total error has an interior minimum, and the linearization residual stays lower order across the tested windows.

B.5 Robustness Summaries

The real-stream, validity-gap, and online-controller sweeps are summarized here in compact form.

Variable	Runs	Interior	Best-window range	Edge/best range
nswprice	9	9/9	14–435	1.31–1.68
nswdemand	9	9/9	25–327	1.15–3.24

Table 10: ELEC2 robustness summary across anchor sizes 24, 48, and 96 and segment starts 0, 8000, and 16000, with segment length 4000. The interior-optimum count records how often the best window stays away from the search boundary; the edge/best range reports the ratio between the worse boundary error and the optimum.

Detector input	Detector	Detection rate	Mean-gap range	Excess-area range
Observation	ADWIN	1.00	204–660	7.23–24.85
Observation	Page–Hinkley	1.00	204–615	6.00–26.17
Observation	KSWIN	0.17–1.00	302–844	3.98–236.31
Observation	CUSUM	1.00	78–232	1.52–6.86
Abs. residual	ADWIN	0.17–1.00	365–903	32.70–277.83
Abs. residual	Page–Hinkley	0.83–1.00	346–969	109.55–188.64
Abs. residual	KSWIN	0.17–0.50	498–1310	2.15–311.09
Abs. residual	CUSUM	0.17–1.00	323–920	20.28–172.90

Table 11: False-alarm-calibrated validity-detection robustness summary on the compact frontier with $H \in \{0.5, 0.75, 1.0\}$, operating windows 160 and 240, roughness levels 0.01 and 0.024, and seeds $0, \dots, 5$. Mean-gap and excess-area ranges are computed over the successful validity-expiry runs in each detector/input family.

Profile	Best static/oracle	Activity/oracle	EWMA/oracle	Kalman/oracle	Plug-in/oracle	Structural/oracle	Adaptive/oracle
Default	1.43	1.36	1.34	1.28	4.41	1.36	1.30
Smooth	1.05	1.41	1.37	1.31	3.24	1.22	1.24
Rough	1.09	1.65	1.64	1.58	2.59	1.30	1.39
Alternating	1.02	1.65	1.60	1.56	2.30	1.24	1.36
Ramp-up	1.02	1.34	1.28	1.21	2.91	1.18	1.23

Table 12: Online memory-selection baselines across five profile families, aggregated over seeds $0, 1, 2$, with maximum one-step window-index jumps. Entries report mean absolute error relative to the oracle controller. The structural controller consistently improves on the raw plug-in rule, beats the adaptive EWMA baseline on 13/15 runs and the adaptive scalar Kalman baseline on 11/15 runs, and is strongest on the rough and alternating profiles.

Scheme	Best window	MAE	n_{eff}	Mean lag	Horizon proxy
Uniform	12	0.272	12.00	5.50	3.880
Triangular	18	0.257	13.86	5.67	3.941
Geometric	15	0.258	12.15	4.95	3.607

Table 13: Weight-sensitivity summary for uniform, triangular, and geometric tapers. The table reports the best window under mean absolute error together with the induced effective sample size, mean lag, and the horizon proxy $n_{\text{eff}}^{-a} + M_H$ used in the diagnostic. Geometric tapering reduces lag most strongly, while triangular tapering enlarges effective sample size at a higher lag cost.

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